

A New Generalized Lattice Attack Against a Family of RSA-Like Cryptosystems

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Abstract

Recently, Cotan and Teşeleanu (NordSec 2023) published an RSA-like cryptosystem. While in RSA, the public exponent e and the private exponent d are related by the equation $ed - k(p - 1)(q - 1) = 1$, in their scheme, e and d are related to the equation $ed - k(p^n - 1)(q^n - 1) = 1$ for some positive integer n . Teşeleanu (CSCML 2024) showed that one can factor the modulus N using a lattice attack if d is small. In this paper, we extend his attack by showing that if the private exponent is either too small or too large, one can factor N in polynomial time by solving the generalized equation $eu - (p^n - 1)(q^n - 1)v = \pm 1$ using lattice reduction techniques.

Keywords: RSA, Crypanalysis, Coppersmith's method, LLL, Lattice attack

1. Introduction

The RSA cryptosystem [22], introduced in 1978 by Rivest, Shamir, and Adleman, is based on the assumed hardness of the integer factorization problem, where $N = pq$ is a product of two large prime numbers of similar size. It is an asymmetric encryption scheme in which the public key (N, e) is used for encryption, while the private key (N, d) enables decryption. Specifically, encryption of a message $m < N$ is performed using modular exponentiation $c \equiv m^e \pmod{N}$, whereas decryption relies on: $m \equiv c^d \pmod{N}$, where d is the modular inverse of e modulo $\varphi(N) = (p - 1)(q - 1)$. The fundamental

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relation between e and d is given by the key equation $ed - k(p-1)(q-1) = 1$ for some integer k .

While RSA is widely adopted for securing communications, extensive cryptanalysis has revealed potential vulnerabilities, including non-factorization-based attacks. Consequently, selecting the parameters carefully, particularly the exponents e and d as well as the primes p and q , is essential to ensure a high level of security against various cryptanalytic attacks. In fact, several attacks on RSA-like cryptosystems leverage the algebraic structure of the key equation [19, 3, 10, 21, 14, 2, 23, 13, 1, 20]. In 1990, Wiener [27] proposed a polynomial time attack that exploits the equation

$$ed - k(p-1)(q-1) = 1$$

to factor N when the private exponent d is sufficiently small, specifically when $d < \frac{1}{3}N^{0.25}$. Over time, Wiener’s attack has been extended through lattice reduction techniques and Coppersmith’s method [19, 15]. In Eurocrypt 1999, Boneh and Durfee [4] refined this approach using the LLL algorithm [17], and improved the bound to $d < N^{0.292}$. In 2023, Cotan and Teşeleanu [8] introduced a family of RSA-like encryption schemes that uses the key equation

$$ed - k(p^n - 1)(q^n - 1) = 1.$$

They showed that d can be recovered in polynomial time regardless of the value of n when $d < N^{0.25n}$. In [26], Teşeleanu improved this bound and factored the module N using a lattice-based method.

In this paper, we focus exclusively on the cryptanalysis of Cotan and Teşeleanu [8] RSA-like cryptosystems. More precisely, we show that if the private exponent is either too small or too large, one can factor N in polynomial time by solving the equation

$$eu - (p^n - 1)(q^n - 1)v = \pm 1$$

using LLL-reduction and Coppersmith’s method [7, 6].

1.1. Related works

In this section, we review the existing attacks on the Cotan and Teşeleanu RSA-like cryptosystems [8].

Cotan and Teşeleanu attack [8]. At NordSec 2023, Cotan and Teşeleanu introduced a new family of RSA-like cryptosystems that uses the key equation

$$ed - k(p^n - 1)(q^n - 1) = 1.$$

The authors also showed that if $e < \varphi_n(N)$ satisfies $ed - k\varphi_n(N) = 1$ with

$$d < \sqrt{\frac{\sqrt{2}^n N^n (\sqrt{N^n} - \delta_n)}{e (\sqrt{2}^n - 1)^2}}, \quad (1)$$

where

$$\varphi_n(N) = (p^n - 1)(q^n - 1) \text{ and } \delta_n = \frac{4\sqrt{2}^n}{(\sqrt{2}^n - 1)^2} + \frac{2(2^n + 1)}{\sqrt{2}^n},$$

then we can recover d in polynomial time. Moreover, if $e \simeq N^n$ and $N \simeq 2^{2\lambda}$, then Equation 1 is equivalent to

$$\log_2(d) < 0.5n\lambda + \frac{n}{4} - \log_2(\sqrt{2}^n - 1) \simeq 0.5n\lambda.$$

Teşeleanu's attack [26]. In 2024, Teşeleanu presented a lattice based small private key attack against the Cotan and Teşeleanu RSA-like cryptosystems. He proved that if $q < p < 2q$ and $d < N^\gamma$, then one can factor N in polynomial time if

$$\begin{cases} \gamma \leq n(1 - \sqrt{0.5}), & \text{when } n = 1 \text{ or } n = 2, \\ \gamma \leq \frac{3n - 1}{4} - \frac{n^2}{2(n + 1)}, & \text{otherwise,} \end{cases}$$

using a lattice based method.

Seck *et al.* attack [24]. In 2025, Seck *et al.* introduced the general equation

$$eu - (p^n - 1)(q^n - 1)v = w$$

with positive integers u, v , and $w \in \mathbb{Z}$ and showed that if d is either small or too large, then N can be factored in polynomial time. More precisely, they proved that if e satisfies $eu - (p^4 - 1)(q^4 - 1)v = w$ and

$$uv < \frac{2N^4 - 49N^2 + 2}{4N + 170N^2} \quad \text{and} \quad |w| < vN,$$

then one can factor N in polynomial time by combining continued fractions with Coppersmith's method. Specifically, if d satisfies

$$d < \sqrt{\frac{2N^4 - 49N^2 + 2}{4N + 170N^2}},$$

then one can factor N in polynomial time.

1.2. Our Contributions

This paper presents an attack against the public key encryption scheme proposed by Cotan and Teşeleanu [8] using lattice reduction techniques and Coppersmith’s method. We study the generalized key equation $eu - \varphi_n(N)v = \pm 1$ with $\varphi_n(N) = (p^n - 1)(q^n - 1)$, and our contributions can be summarized as follows:

- First, we prove that for even integers n , $\varphi_n(N)$ can be expressed as a polynomial in $p - q$, where the coefficients depend only on N and n , i.e., $\varphi_n(N) = -(p - q)^n - \sum_{i=0}^{n-1} a_i(p - q)^i$. For odd integers n , a similar result was previously established by Teşeleanu [26].
- Next, we transform the key equation $eu - \varphi_n(N)v = \pm 1$ into a modular equation of the form $xh(y) + c \equiv 0 \pmod{e}$, where $h(y) = y^n + \sum_{i=0}^{n-1} a_i y^i$ and $(v, p - q)$ or $(v, p + q)$ constitutes a small solution. Using a result from Kunihiro [16] for solving bivariate modular equations, along with a recent result from Seck *et al.* [24], we demonstrate that the modulus N can be factored in polynomial time using Coppersmith’s strategy if the private exponent d is either too large (e.g., $d = \varphi_n(N) - N^\delta$) or too small (e.g., $d = N^\delta$), under certain conditions on δ .
- Finally, we provide a proof-of-concept implementation¹ in SageMath [11] to illustrate the effectiveness of our attack when d is too large. We include a numerical example along with an experimental performance of our attack for different parameters.

Organization of this paper. The rest of this paper is structured as follows. In Section 2, we recall the different algorithms of Cotan and Teşeleanu cryptosystem. We also provide background on lattices and the method for finding small roots of polynomials. Section 3 is devoted to the main results of this paper. We show under which conditions on the private exponent d , one can factor a modulus N in polynomial time given a public key (N, e) . In Section 4, we present a numerical example of our attack, along with a performance evaluation for different parameters. We compare our attack with existing ones based on either lattice techniques or continued fractions in Section 5. Finally, we conclude in Section 6.

¹<https://github.com/mseckept/generalized-lattice-attack.git>

2. Preliminaries

In this section, we first summarize the RSA-like cryptosystem of Cotan and Teşeleanu. Next, we review some useful properties of lattices and the connection between lattice reduction techniques and strategies for solving modular polynomials. Finally, we end this section by recalling key lemmas and theorems relevant to our attack.

2.1. Cotan and Teşeleanu Scheme

In NordSec 2023, Cotan and Teşeleanu [8] introduced a new RSA-like cryptosystem using the key equation

$$ex - k(p^n - 1)(q^n - 1) = 1 \text{ for some positive integer } n.$$

Notice that the previous equation generalizes the standard RSA equation [22] ($n = 1$) as well as the one from the scheme by Elkamchouchi, Elshenawy and Shaban scheme ($n = 2$) proposed in 2002 [12].

Let

$$\mathbb{A}_n(p) = \frac{\mathbb{Z}/p\mathbb{Z}[t]}{t^n - r},$$

where p is a prime number, n a positive integer and $r \in \left(\frac{\mathbb{Z}}{p\mathbb{Z}}\right)^\star$ such that $t^n - r$ is an irreducible polynomial over $\frac{\mathbb{Z}}{p\mathbb{Z}}[t]$. Notice that $t^n - r$ is irreducible over $\frac{\mathbb{Z}}{p\mathbb{Z}}[t]$ if and only if the following two conditions hold: (i) each prime factor p_i of n divides the order e of r in $\left(\frac{\mathbb{Z}}{p\mathbb{Z}}\right)^\star$, but not $(p-1)/e$; (ii) $p \equiv 1 \pmod{4}$ if $n \equiv 0 \pmod{4}$ (cf. Theorem 3.75, p. 124 in [18]).

Since $\mathbb{A}_n(p) = GF(p^n)$ is a finite field of order p^n , it can be represented as follows:

$$\mathbb{A}_n(p) = \left\{ a(t) = a_0 + a_1t + \dots + a_{n-1}t^{n-1} : a_0, a_1, \dots, a_{n-1} \in \frac{\mathbb{Z}}{p\mathbb{Z}} \right\}$$

Let $a(t), b(t) \in \mathbb{A}_n(p)$, then the quotient field induces a natural product

◦

$$\begin{aligned}
a(t) \circ b(t) &= \left(\sum_{i=0}^{n-1} a_i t^i \right) \circ \left(\sum_{j=0}^{n-1} b_j t^j \right) \\
&= \sum_{i=0}^{n-2} \left(\sum_{j=0}^i a_j b_{i-j} + r \sum_{j=0}^{i+n} a_j b_{i-j+n} \right) t^i + \sum_{j=0}^{b-1} a_j b_{n-1-j} t^{n-1}.
\end{aligned}$$

Moreover, $\mathbb{A}_n^*(p)$, the set of invertible elements in $\mathbb{A}_n(p)$, is a cyclic group of order $\varphi_n(p) = p^n - 1$. Let $a(t) \in \mathbb{A}_n^*(p)$ and e be an integer, we define

$$[a(t)]^e = a(t) \circ a(t) \circ \dots \circ a(t) \text{ (} e \text{ times)}.$$

The key generation, encryption and decryption algorithms are very similar to the RSA cryptosystem and are given as follows.

Key Generation : Let λ be a security parameter and $n \geq 1$ an integer.

- Randomly generate two distinct large primes p, q such that $p, q \geq 2^\lambda$ and compute $N = pq$.
- Choose an integer $r \in \mathbb{Z}/N\mathbb{Z}$ such that $t^n - r$ is an irreducible polynomial over $\frac{\mathbb{Z}}{p\mathbb{Z}}[t]$ and $\frac{\mathbb{Z}}{q\mathbb{Z}}[t]$. Let $\varphi_n(N) = (p^n - 1)(q^n - 1)$.
- Select a positive integer $e < \varphi_n(N)$, with $\gcd(e, \varphi_n(N)) = 1$.
- Compute $d = e^{-1} \pmod{\varphi_n(N)}$. The public key is $pk = (N, e, n, r)$ and the private key is $sk = (p, q, d)$.

Encryption process : To encrypt a message $m = (m_0, m_1, \dots, m_{n-1}) \in (\mathbb{Z}/N\mathbb{Z})^n$ with the public key $pk = (N, e, n, r)$,

- Represent the message m as $m(t) = m_0 + m_1 t + \dots + m_{n-1} t^{n-1} \in \mathbb{A}_n^*(N)$.
- The ciphertext is $c(t) = [m(t)]^e \pmod{N}$.

Decryption process : To decrypt a ciphertext $c(t) \in \mathbb{A}_n^*(N)$ with the private key $sk = (p, q, d)$, compute

$$m(t) = [c(t)]^d \pmod{N}$$

Notice that one can use the Chinese Remainder Theorem (CRT) to compute the value $[c(t)]^d \pmod{N}$.

Correctness : We have

$$\begin{aligned}
|\mathbb{A}_n^*(N)| &= \left| \left(\frac{\mathbb{Z}/(pq)\mathbb{Z}[t]}{t^n - r} \right)^* \right| \\
&= \left| \left(\frac{\mathbb{Z}/p\mathbb{Z}[t]}{t^n - r} \right)^* \times \left(\frac{\mathbb{Z}/q\mathbb{Z}[t]}{t^n - r} \right)^* \right| \\
&= |\mathbb{A}_n^*(p)| \times |\mathbb{A}_n^*(q)| \\
&= (p^n - 1)(q^n - 1) = \varphi_n(N)
\end{aligned}$$

This show that, for any $[a(t)] \in \mathbb{A}_n^*(N)$, the following holds:

$$[a(t)]^{(p^n-1)(q^n-1)} \equiv 1 \pmod{N} \quad (2)$$

Therefore $([m(t)]^e)^d \pmod{N} = [m(t)]$ for all $[m(t)] \in \mathbb{A}_n^*(N)$ since $ed = 1 + k\varphi_n(N)$ for some integer k .

2.2. Lattices and Small Roots of Polynomials

Let m, ω be two positive integers. Let $\{b_1, b_2, \dots, b_\omega\}$ be a set of ω linearly independent vectors in $\mathbb{R}^m$. The lattice generated by $\{b_1, b_2, \dots, b_\omega\}$ [25, 9] is given by

$$\mathcal{L}(b_1, b_2, \dots, b_\omega) = \left\{ \sum_{i=1}^{\omega} a_i b_i : a_i \in \mathbb{Z} \right\},$$

which is the set of all integer linear combinations of the vectors $b_1, b_2, \dots, b_\omega$. The integer ω is called the rank of the lattice, and if $\omega = m$, the lattice is called a full-rank lattice. The basis of $\mathcal{L}$ can be represented as a matrix B , called a basis matrix, whose rows are the basis vectors $b_1, b_2, \dots, b_\omega$, i.e.,

$$B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_\omega \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{\omega 1} & b_{\omega 2} & \cdots & b_{\omega m} \end{pmatrix}.$$

In this case,

$$\mathcal{L}(B) = \{aB : a \in \mathbb{Z}^\omega\}$$

and B is a square matrix if $\mathcal{L}(B)$ is a full-rank lattice. The determinant of $\mathcal{L}(B)$ is given by $\det(\mathcal{L}) := \sqrt{\det(BB^T)}$. An important computational

problem in lattices is the γ -Approximate Shortest Vector Problem (γ -SVP). Given a basis matrix B of a lattice $\mathcal{L}$ and an approximation factor $\gamma \geq 1$, the γ -SVP consist in finding a nonzero lattice vector $v \in \mathcal{L}$ such that:

$$\|v\| \leq \gamma \cdot \lambda_1(\mathcal{L}),$$

where

$$\lambda_1(\mathcal{L}) := \min_{y \in \mathcal{L} \setminus \{0\}} \|y\|$$

is the length of the shortest nonzero vector in $\mathcal{L}$, and $\|\cdot\|$ denotes the Euclidean (ℓ_2) norm. When $\gamma = 1$, we have the exact SVP, which is known to be NP-hard under randomized reductions. For $\gamma = 2^{O(n)}$, the well-known LLL reduction algorithm [17] can solve the γ -SVP in polynomial time. The LLL algorithm yields a reduced basis $\{v_1, v_2, \dots, v_\omega\}$ of a given ω -dimensional lattice $\mathcal{L}(B)$, satisfying the property:

$$\|v_i\| \leq 2^{\frac{\omega(\omega-1)}{4(\omega+1-i)}} \cdot \det(\mathcal{L})^{\frac{1}{\omega+1-i}}, \quad 1 \leq i \leq \omega.$$

The LLL algorithm finds many applications in cryptanalysis, particularly in finding small roots of modular equations. Howgrave-Graham [15, 19] established in 1997 an essential lemma to determine whether a root of a modular polynomial equation also corresponds to a root of a polynomial over the integers.

Lemma 1 (Howgrave-Graham). *Let $g(x_1, \dots, x_f) \in \mathbb{Z}[x_1, \dots, x_f]$ be an integer polynomial consisting of at most ω monomials. Let $m, e, X_1, \dots, X_f$ be known positive integers. If the following two conditions are satisfied:*

- $g(x_1^0, \dots, x_f^0) \equiv 0 \pmod{e^m}$, where $|x_1^0| \leq X_1, \dots, |x_f^0| \leq X_f$,
- $\|g(x_1 X_1, \dots, x_f X_f)\| < \frac{e^m}{\sqrt{\omega}}$,

then it follows that $g(x_1^0, \dots, x_f^0) = 0$ holds over the integers.

2.3. Useful Theorems and Lemmas

We begin with an important result due to Kunihiro [16] about the solutions of bivariate modular polynomial equations. Specifically, he has proposed an efficient algorithm based on lattices to find a small root (x_0, y_0) of equation $f(x, y) \equiv 0 \pmod{e}$ with $f(x, y) = xH(y) + c \in \mathbb{Z}[x, y]$ for some constant c , under certain conditions on x_0 and y_0 .

Theorem 2 ([16, 26]). *Let $H(y) \in \mathbb{Z}[y]$ be a monic polynomial of degree $r \geq 1$ and e be an integer. Suppose that*

- $x_0 H(y_0) + c \equiv 0 \pmod{e}$ for some $|x_0| < X = e^{\delta'}$, $|y_0| < Y = e^{\gamma'}$;
- $|c| < XY^r$.

Then one can solve the equation $xH(y) + c \equiv 0 \pmod{e}$ if

$$\begin{cases} \delta' \leq \frac{r+2}{2(r+1)} - \frac{r+1}{2}\gamma', & \text{when } 0 < \gamma' < \frac{r}{(r+1)^2}, \\ \delta' \leq 1 - \sqrt{r\gamma'}, & \text{when } \frac{r}{(r+1)^2} \leq \gamma' \leq \frac{1}{r}. \end{cases}$$

For the sake of completeness, we give, in the following, the different steps of Kunihiro's strategy [16] to recover small roots (x_0, y_0) of the modular equation $xH(y) + c \equiv 0 \pmod{e}$ with

$$|x_0| < X = e^{\delta'}, |y_0| < Y = e^{\gamma'}$$

To solve the modular equation $xH(y) + c \equiv 0 \pmod{e}$ with

$$H(y) = y^r + \sum_{i=0}^{r-1} a_i y^i,$$

Kunihiro [16] applies Howgrave-Graham's Lemma 1 and proceeds as follows.

- Define a new variable $z = xy^r + c$. In this case the modular equation become $z + xh(y) \equiv 0 \pmod{e}$ with $h(y) = \sum_{i=0}^{r-1} a_i y^i$.
- Define the shift-polynomials

$$g_{[i,j,k]}(x, y, z) = x^i y^j (z + xh(y))^k e^{m-k}, \text{ where } m \text{ is a positive integer}$$

Notice that all monomials xy^r should be replaced by $z - c$.

- Consider the set of shift-polynomials $\mathcal{F} = \mathcal{G} \cup \mathcal{H}$ with

$$\begin{aligned} \mathcal{G} &= \{g_{[u-i,j,i]} : u = 0, \dots, m; i = 0, \dots, u; j = 0, \dots, r-1\}, \\ \mathcal{H} &= \{g_{[0,j,u]} : u = 0, \dots, m; j = r, \dots, r + \lfloor \tau u \rfloor - 1\}. \end{aligned}$$

where τ is a parameter s.t. $0 \leq \tau \leq 1$.

- Let $Z = XY^r + |c|$. Define the lattice basis matrix $B(m, \tau)$ where all its rows are coefficient vectors of shift-polynomials $g_{[i,j,k]}(xX, yY, zZ)$, according to the following ordering in $\mathcal{F}$:

$$g_{[i,j,k]} <_p g_{[i',j',k']} \iff \begin{cases} i + k < i' + k' \text{ or} \\ i + k = i' + k' \text{ and } i < i' \text{ or} \\ i = i', k = k' \text{ and } j \leq j'. \end{cases}$$

And

$$x^i y^j z^k <_m x^{i'} y^{j'} z^{k'} \iff \begin{cases} i + k < i' + k' \text{ or} \\ i + k = i' + k' \text{ and } i < i' \text{ or} \\ i = i', k = k' \text{ and } j \leq j'. \end{cases}$$

Notice that the orderings $<_p$ and $<_m$ ensure that the lattice basis matrix $B(m, \tau)$ is triangular.

- Apply the LLL reduction algorithm on $B(m, \tau)$. Let $v_i, i \in \mathcal{I}$ be the shortest vectors in the LLL-reduced basis. Construct the polynomials $f_i(x, y, z)$ from v_i .
- Solve the system $f_i(x, y, z) = 0$ for $i \in \mathcal{I}$ over integers using the Gröbner bases method or the resultants method.

Notice that Kunihiro's algorithm is heuristic and is based in the following assumption: the reduced polynomials $f_i(x, y, z), i \in \mathcal{I}$ generated by the LLL algorithm are algebraically independent. Consequently, we assume in this paper that this assumption holds.

Recently, Seck *et al.* [24] showed that if a private exponent $d = N^\delta$ of an RSA-like cryptosystem is too large, then one can factor the modulus N , under certain conditions on δ .

Theorem 3 ([24]). *Let $N = pq$ be an RSA modulus. Let $\psi : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Suppose $\mathcal{A}$ is an algorithm that is able to factor N in polynomial time given N and a public exponent $0 < e < \psi(p, q)$ such that there exist positive integers x and y with $x < \mathcal{B}_1 \in \mathbb{R}^+$ satisfying*

$$ex - y\psi(p, q) = z \text{ with } |z| \leq \mathcal{B}_2, \mathcal{B}_2 \geq 1.$$

Then, using $\mathcal{A}$, one can factor N in polynomial time given N and a public exponent $0 < e' < \psi(p, q)$ such that the corresponding private exponent $d' = \psi(p, q) - d$ for some $d < \mathcal{B}_1$.

For a balanced RSA modulus $N = pq$ with $p > q$, the sum $p + q$ and the difference $p - q$ can be bounded as in the following Lemma 4.

Lemma 4. [5] *Let $N = pq$ be an RSA modulus with $q < p < 2q$. The following holds:*

$$2\sqrt{N} < p + q < 3\sqrt{N} \text{ and } 0 < p - q < \sqrt{N}$$

We can assume without loss of generality, in the rest of this paper, that $p - q > N^{\frac{1}{4}}$; otherwise N can be factored in polynomial time using the Fermat factorization technique [19].

3. Our New Generalized Lattice Attack

In this section, we detail our lattice attack on the generalized equation

$$eu - (p^n - 1)(q^n - 1)v = \pm 1$$

for some positive integer n . Before explaining our attack, we will prove the following lemma.

Lemma 5. *Let $N = pq$ be a product of two primes. Without loss of generality (Wlog), we can suppose that $q < p < \mu q$ for some positive real μ . Then the following holds:*

$$N^n + 1 - ((\mu N)^{\frac{n}{2}} + N^{\frac{n}{2}}) < \varphi_n(N) < N^n + 1 - \left(\left(\frac{N}{\mu} \right)^{\frac{n}{2}} + N^{\frac{n}{2}} \right).$$

Proof. In the one hand, we have $q < p < \mu q$, then multiplying by p we get $N < p^2 < \mu N$. Then $\sqrt{N} < p < \sqrt{\mu N}$ which implies that

$$N^{\frac{n}{2}} < p^n < (\mu N)^{\frac{n}{2}}. \quad (3)$$

In the other hand, since $p = N/q$, we have $\sqrt{N} < N/q < \sqrt{\mu N}$. Then $\sqrt{\frac{N}{\mu}} < q < \sqrt{N}$, therefore

$$\left(\frac{N}{\mu} \right)^{\frac{n}{2}} < q^n < N^{\frac{n}{2}}. \quad (4)$$

By combining Eq. 3 and 4, we have

$$N^{\frac{n}{2}} + \left(\frac{N}{\mu}\right)^{\frac{n}{2}} < p^n + q^n < (\mu N)^{\frac{n}{2}} + N^{\frac{n}{2}}.$$

This shows that

$$N^n + 1 - ((\mu N)^{\frac{n}{2}} + N^{\frac{n}{2}}) < \varphi_n(N) < N^n + 1 - \left(\left(\frac{N}{\mu}\right)^{\frac{n}{2}} + N^{\frac{n}{2}}\right).$$

□

Let us now describe our attack in detail. We distinguish two cases: when n is even and when n is odd.

3.1. Case I: The New Attack when n is even

Suppose $n = 2k$ for some positive integer k . We have the following.

Lemma 6. *Let $N = pq$ be an RSA modulus. The function $\varphi_n(N) = (p^n - 1)(q^n - 1)$ can be expressed as a polynomial in $p - q$, where the coefficients depend only on N and n . That is, there exist coefficients $a_0, a_1, \dots, a_{n-1}$ depending only on N and n such that*

$$\varphi_n(N) = -(p - q)^n - \sum_{i=0}^{n-1} a_i (p - q)^i.$$

Furthermore, $a_j = 0$ for odd j ($j = 1, 3, 5, \dots, n - 1$) i.e.

$$\varphi_n(N) = -(p - q)^n - \sum_{j=0}^{\frac{n-2}{2}} a_j (p - q)^{2j}.$$

Proof. We have $\varphi_n(N) = (p^n - 1)(q^n - 1) = N^n - (p^n + q^n) + 1$. Let us now write $p^n + q^n$ as a polynomial in $p - q$.

$$\begin{aligned} (p - q)^n &= \sum_{i=0}^n \binom{n}{i} p^{n-i} (-q)^i \\ &= p^n + q^n + \sum_{i=1}^{n-1} \binom{n}{i} p^{n-i} (-q)^i. \end{aligned}$$

Then we have:

$$\begin{aligned}
p^n + q^n &= (p - q)^n - \sum_{i=1}^{n-1} \binom{n}{i} p^{n-i} (-q)^i \\
&= (p - q)^n - (-1)^k \binom{n}{\frac{n}{2}} N^{\frac{n}{2}} - \sum_{j=1}^{\frac{n}{2}-1} \left[\binom{n}{j} p^{n-j} (-q)^j + \binom{n}{n-j} p^j (-q)^{n-j} \right] \\
&= (p - q)^n - (-1)^k \binom{n}{\frac{n}{2}} N^{\frac{n}{2}} - \sum_{j=1}^{\frac{n}{2}-1} (-1)^j \left[\binom{n}{j} p^{n-j} q^j + \binom{n}{j} p^j q^{n-j} \right] \\
&= (p - q)^n - (-1)^k \binom{n}{\frac{n}{2}} N^{\frac{n}{2}} - \sum_{j=1}^{\frac{n}{2}-1} (-1)^j \binom{n}{j} N^j (p^{n-2j} + q^{n-2j}).
\end{aligned}$$

This show that

$$\begin{aligned}
\varphi_n(N) &= -(p - q)^n + \sum_{j=1}^{\frac{n}{2}-1} \left[(-1)^j \binom{n}{j} N^j (N^{n-2j} + 1 - \varphi_{n-2j}(N)) \right] \\
&\quad + (-1)^{\frac{n}{2}} \binom{n}{\frac{n}{2}} N^{\frac{n}{2}} + N^n + 1.
\end{aligned} \tag{5}$$

For $n = 2$, we have $\varphi_2(N) = -(p - q)^2 - 2N + N^2 + 1$. Therefore, by induction, we get the result using the recurrence relation in Equation 5. Notice that all monomials in $\varphi_{n-2j}(N)$ are even, Therefore $\varphi_n(N)$ can be written in the form

$$\varphi_n(N) = -(p - q)^n - \sum_{j=0}^{\frac{n-2}{2}} a_j (p - q)^{2j}.$$

□

Using the formula in Equation 5, we compute $\varphi_n(N)$, $n = 2, 4, 6, 8, 10$.

$$\begin{aligned}
\varphi_2(N) &= -(p - q)^2 + (N - 1)^2 \\
\varphi_4(N) &= -(p - q)^4 - 4N(p - q)^2 + (N^2 - 1)^2 \\
\varphi_6(N) &= -(p - q)^6 - 6N(p - q)^4 - 9N^2(p - q)^2 + (N^3 - 1)^2 \\
\varphi_8(N) &= -(p - q)^8 - 8N(p - q)^6 - 20N^2(p - q)^4 - 16N^3(p - q)^2 + (N^4 - 1)^2 \\
\varphi_{10}(N) &= -(p - q)^{10} - 10N(p - q)^8 - 35N^2(p - q)^6 - 50N^3(p - q)^4 + \\
&\quad - 25N^4(p - q)^2 + (N^5 - 1)^2
\end{aligned}$$

Theorem 7. Let $N = pq$ be a product of two unknown primes with $q < p < \mu q$ and n a positive integer. Let $\gamma = \log(s)/\log(N)$ where s is one of the smallest positive integers such that $\frac{N^n}{s} < N^n + 1 - ((\mu N)^{\frac{n}{2}} + N^{\frac{n}{2}})$. Let $e = N^\beta$, $u < N^\delta$ and $p - q < N^\theta$. Suppose that the public exponent e satisfies the generalized equation

$$eu - (p^n - 1)(q^n - 1)v = w \text{ with } w = \pm 1.$$

Then N can be factored in polynomial time if

$$\begin{cases} \delta \leq n - \sqrt{\beta n \theta} - \gamma, & \text{when } n\theta \leq \beta \leq \frac{(n+1)^2 \theta}{n} \\ \delta \leq n - \frac{n\beta}{2(n+1)} - \frac{(n+1)\theta}{2} - \gamma, & \text{when } \frac{(n+1)^2 \theta}{n} \leq \beta \leq \frac{(2n - (n+1)\theta - 2\gamma)(n+1)}{n}. \end{cases}$$

Proof. Assuming the public exponent $e = N^\beta$ satisfies the generalized equation $eu - (p^n - 1)(q^n - 1)v = w$ with $u < N^\delta$, $p - q < N^\theta$ and $w = \pm 1$. Assuming also that the following holds

$$\begin{cases} \delta \leq n - \sqrt{\beta n \theta} - \gamma, & \text{when } n\theta \leq \beta \leq \frac{(n+1)^2 \theta}{n} \\ \delta \leq n - \frac{n\beta}{2(n+1)} - \frac{(n+1)\theta}{2} - \gamma, & \text{when } \frac{(n+1)^2 \theta}{n} \leq \beta \leq \frac{(2n - (n+1)\theta - 2\gamma)(n+1)}{n}. \end{cases}$$

We have

$$v = \frac{eu - w}{(p^n - 1)(q^n - 1)} \leq \frac{eu + |w|}{(p^n - 1)(q^n - 1)} \leq \frac{N^{\beta+\delta} + 1}{(p^n - 1)(q^n - 1)}.$$

We have $\varphi_n(N) = (p^n - 1)(q^n - 1) = N^n - (p^n + q^n) + 1$. Clearly $\varphi_n(N) < N^n$, but by Lemma 5, we have

$$N^n + 1 - ((\mu N)^{\frac{n}{2}} + N^{\frac{n}{2}}) < \varphi_n(N)$$

Let s be one of the smallest positive integers such that

$$\frac{N^n}{s} < N^n + 1 - ((\mu N)^{\frac{n}{2}} + N^{\frac{n}{2}}) < \varphi_n(N) < N^n \quad (6)$$

This implies that $\frac{1}{\varphi_n(N)} < \frac{s}{N^n}$. Therefore

$$v < sN^{\beta+\delta-n}, u < N^\delta, e = N^\beta \text{ and } (p - q) < N^\theta.$$

Moreover, the generalized key equation $eu - (p^n - 1)(q^n - 1)v = w$ can be transformed into the modular form

$$-(p^n - 1)(q^n - 1)v - w \equiv 0 \pmod{e}.$$

By Lemma 6, $\varphi_n(N) = -(p - q)^n - \sum_{i=0}^{\frac{n-2}{2}} a_i(p - q)^{2i}$ for some $a_i, i = 0, \dots, \frac{n-2}{2}$ depending only on N and n . Therefore

$$\left((p - q)^n + \sum_{i=0}^{\frac{n-2}{2}} a_i(p - q)^{2i} \right) v + (-w) \equiv 0 \pmod{e}.$$

Define the following function $f(x, y) = xh(y) + c$ with $c = -w = \pm 1$ and

$$h(y) = y^n + \sum_{i=0}^{\frac{n-2}{2}} a_i y^{2i}.$$

Then $(v, \pm(p - q))$ are roots of the modular polynomial equation $f(x, y) \equiv 0 \pmod{e}$. We set the bounds for (x, y) as follows.

$$X = sN^{\beta+\delta-n} = N^{\beta+\delta+\gamma-n} = e^{\frac{\beta+\delta+\gamma-n}{\beta}}, \quad \text{with } \gamma = \frac{\log(s)}{\log(N)}$$

$$Y = N^\theta = e^{\frac{\theta}{\beta}}.$$

Then using Theorem 2, one can find the roots $(v, \pm(p - q))$ of the polynomial $f(x, y)$ if the following holds:

$$\frac{\beta + \delta + \gamma - n}{\beta} \leq \frac{n + 2}{2(n + 1)} - \frac{n + 1}{2} \frac{\theta}{\beta}, \quad \text{when } 0 < \frac{\theta}{\beta} < \frac{n}{(n + 1)^2}, \quad (7)$$

And

$$\frac{\beta + \delta + \gamma - n}{\beta} \leq 1 - \sqrt{\frac{n\theta}{\beta}}, \quad \text{when } \frac{n}{(n + 1)^2} \leq \frac{\theta}{\beta} \leq \frac{1}{n}. \quad (8)$$

From Inequations 7, we have

$$\begin{aligned}
\frac{\beta + \delta + \gamma - n}{\beta} &\leq \frac{n+2}{2(n+1)} - \frac{n+1}{2} \frac{\theta}{\beta} \\
&\iff \beta + \delta + \gamma - n \leq \frac{n+2}{2(n+1)}\beta - \frac{n+1}{2}\theta \\
&\iff \delta \leq \frac{n+2}{2(n+1)}\beta - \frac{n+1}{2}\theta - \beta - \gamma + n \\
&\iff \delta \leq n - \frac{n\beta}{2(n+1)} - \frac{(n+1)\theta}{2} - \gamma.
\end{aligned}$$

On the one hand, we want $\delta \geq 0$, which implies that

$$\begin{aligned}
n - \frac{n\beta}{2(n+1)} - \frac{(n+1)\theta}{2} - \gamma \geq 0 &\iff \frac{n\beta}{2(n+1)} \leq \frac{2n - (n+1)\theta - 2\gamma}{2} \\
&\iff \beta \leq \frac{(2n - (n+1)\theta - 2\gamma)(n+1)}{n}. \quad (9)
\end{aligned}$$

On the other hand, from Inequality (7), we have

$$0 < \frac{\theta}{\beta} < \frac{n}{(n+1)^2} \implies \frac{(n+1)^2\theta}{n} \leq \beta. \quad (10)$$

From Inequalities (10) and (9), we obtain the following bound for β :

$$\frac{(n+1)^2\theta}{n} \leq \beta \leq \frac{(2n - (n+1)\theta - 2\gamma)(n+1)}{n}. \quad (11)$$

Now, from Inequations 8, we have

$$\begin{aligned}
\frac{\beta + \delta + \gamma - n}{\beta} \leq 1 - \sqrt{\frac{n\theta}{\beta}} &\iff \beta + \delta + \gamma - n \leq \beta - \sqrt{\beta n \theta} \\
&\iff \delta \leq n - \sqrt{\beta n \theta} - \gamma.
\end{aligned}$$

Since we also want $\delta \geq 0 \iff n - \sqrt{\beta n \theta} - \gamma \geq 0 \iff \beta \leq \frac{(n - \gamma)^2}{n}$. By the Inequations 8, β satisfies $\frac{n}{(n+1)^2} \leq \frac{\theta}{\beta} \leq \frac{1}{n}$ which implies that

$$n\theta \leq \beta \leq \frac{(n+1)^2\theta}{n}.$$

Once we get the small roots (x_0, y_0) satisfying $f(x_0, y_0) \equiv 0 \pmod{e}$, we can compute the factors p and q of N by solving the following system.

$$\begin{cases} pq = N \\ p - q = |y_0| \end{cases}$$

□

3.2. Case II: The New Attack when n is odd

In this section, we suppose $n = 2k + 1$ for some positive integer k . It is shown in [26] that $\varphi_n(N) = (p^n - 1)(q^n - 1)$ can be expressed as a polynomial in $p + q$, where the coefficients depend only on N . In other words, there exist coefficients $a_i, i = 0, \dots, n - 1$ such that

$$\varphi_n(N) = -(p + q)^n - \sum_{i=0}^{n-1} a_i (p + q)^i.$$

As in the case n even, the following holds.

Theorem 8. *Let $N = pq$ be a product of two unknown primes with $q < p < \mu q$ and n be a positive odd integer. Let $\gamma = \log(s)/\log(N)$ where s is defined as in Equation 6. Let $e = N^\beta$, $u < N^\delta$ and $p + q < N^\theta$. Suppose that the public exponent e satisfies the generalized equation*

$$eu - (p^n - 1)(q^n - 1)v = w \text{ with } w = \pm 1.$$

Then N can be factored in polynomial time if

$$\begin{cases} \delta \leq n - \sqrt{\beta n \theta} - \gamma, & \text{when } n\theta \leq \beta \leq \frac{(n+1)^2 \theta}{n} \\ \delta \leq n - \frac{n\beta}{2(n+1)} - \frac{(n+1)\theta}{2} - \gamma, & \text{when } \frac{(n+1)^2 \theta}{n} \leq \beta \leq \frac{(2n - (n+1)\theta - 2\gamma)(n+1)}{n}. \end{cases}$$

Proof. The proof is similar to the case n even. Just replace $p - q$ by $p + q$. □

Corollary 9. *Let $N = pq$ be a product of two unknown primes with $q < p < \mu q$ and n a positive integer. Let $\gamma = \log(s)/\log(N)$ where s is defined as in Equation 6. Let the public exponent $e = N^\beta$, and $p + q < N^\theta$ (resp. $p - q < N^\theta$) if n is odd (resp. n is even). If the private exponent d satisfies*

$$d < N^\delta \quad \text{or} \quad |(p^n - 1)(q^n - 1) - d| < N^\delta.$$

Then N can be factored in polynomial time if

$$\begin{cases} \delta \leq n - \sqrt{\beta n \theta} - \gamma, & \text{when } n\theta \leq \beta \leq \frac{(n+1)^2 \theta}{n} \\ \delta \leq n - \frac{n\beta}{2(n+1)} - \frac{(n+1)\theta}{2} - \gamma, & \text{when } \frac{(n+1)^2 \theta}{n} \leq \beta \leq \frac{(2n-(n+1)\theta-2\gamma)(n+1)}{n}. \end{cases}$$

Proof. • Suppose that $d < N^\delta$. We know that the public exponent e and the private exponent d are related by the equation $ed - k\varphi_n(N) = 1$. By Theorem 7 and Theorem 8, we obtain the desired result.

• Now suppose that

$$|(p^n - 1)(q^n - 1) - d| < N^\delta \iff d > (p^n - 1)(q^n - 1) - N^\delta.$$

This means that d can be expressed as $d = \varphi_n(N) - d'$ for some $d' < N^\delta$. Then, by Theorem 3 with $\mathcal{B}_2 = 1$ and $\mathcal{B}_1 = N^\delta$, and using Theorems 7 and 8, one can factor N in polynomial time if

$$\begin{cases} \delta \leq n - \sqrt{\beta n \theta} - \gamma, & \text{if } n\theta \leq \beta \leq \frac{(n+1)^2 \theta}{n}, \\ \delta \leq n - \frac{n\beta}{2(n+1)} - \frac{(n+1)\theta}{2} - \gamma, & \text{if } \frac{(n+1)^2 \theta}{n} \leq \beta \leq \frac{(2n-(n+1)\theta-2\gamma)(n+1)}{n}. \end{cases}$$

□

4. Experimental Results

We have checked the effectiveness of our generalized lattice attack against the Cotan and Teşeleanu scheme [8]. Our experimental environment is Ubuntu 24.04.1 LTS (GNU/Linux 5.15.153.1-microsoft-standard-WSL2 x86_64) on Intel(R) Core(TM) i7-150U 1.8 GHz with 16 GB RAM. A proof-of-concept implementation² with Sagemath [11] 10.4 is provided. We also provide, in this section, a numerical example to explain how our attack works step by step, and present a performance experiment for different parameters.

4.1. A Numerical Example

For this numerical example, we set $n = 4$ i.e $\varphi_n(N) = (p^4 - 1)(q^4 - 1)$ and $\mu = 2$, and wlog we suppose that the private exponent d is too large.

²<https://github.com/mseckept/generalized-lattice-attack.git>

Let us now consider a 300 bitsize modulus N and the public exponent e :

$$\begin{aligned}
N &= 125388489491686550067302101618841769127499464493106944683004004 \\
&\quad 3567660931938711310794106381 \\
e &= 2430080287318190046644932414103595481991229787711287457767417355760 \\
&\quad 8055126231127132165749421957948092456043755464628085616048296961988 \\
&\quad 9366548181622065954369013987345457864977011519407406020607349964317 \\
&\quad 1364965220095238018118838859981753677526439817453413038583853167511 \\
&\quad 1092569979188509756051676128206156756427093589944266198753727605023 \\
&\quad 10158328807521050412943909
\end{aligned}$$

Therefore, we have $\beta = \log(e)/\log(N) \approx 4$ for $e = N^\beta$. Now we have to solve the equation $eu - (p^4 - 1)(q^4 - 1)v = -1$. To do so, we will solve the modular equation $-(p^4 - 1)(q^4 - 1)v + 1 \equiv 0 \pmod{e} \iff f(x, y) = xy^4 + (4N)xy^2 - (N^2 - 1)^2x + 1 \equiv 0 \pmod{e}$ where $x = u$ and $y = p - q$. Now our goal is to find one root (x_0, y_0) of the modular equation $f(x, y) \equiv 0 \pmod{e}$. To launch our lattice attack, we take the following parameters: $\theta = 1/2, s = 6$ (note that the value of s satisfies Inequal. 6), $\delta = 0.9$ and $w = -1$ i.e. the private exponent is large. We get the following bounds:

$$\begin{aligned}
X &= \left\lfloor \frac{6eN^\delta}{N^n} \right\rfloor = 7068826573978490903644620231340428749786 \\
&\quad 956886962571707789244358631326408
\end{aligned}$$

$$Y = \lfloor N^{1/2} \rfloor = 1119770018761381868010203621675156586578699818$$

Next, for the construction of the lattice basis $B(m, \tau)$, we choose $m = 3, \tau = 1$ and get a lattice of dimension $\omega = 46$. Afterwards, we apply the LLL lattice reduction algorithm to get a newly reduced basis and we obtain two integer polynomials. Using the Gröbner basis computation, we obtain the following roots in about 0.1 second.

$$\begin{aligned}
x_0 &= 1424036147889738651116719863674942899327447076 \\
&\quad 945715275467080165445196269 \\
|y_0| &= 270195271935730750938376055531968899992397300
\end{aligned}$$

After that, we solve the following system.

$$\begin{cases}
pq = N = 1253884894916865500673021016188417691274994644 \\
\quad 931069446830040043567660931938711310794106381 \\
p - q = |y_0| = 270195271935730750938376055531968899992397300
\end{cases}$$

We get the following values for p and q .

$$\begin{cases} p = 1262987819523414977122958994742236637808098291 \\ q = 992792547587684226184582939210267737815700991 \end{cases}$$

We get the following private exponent

$$\begin{aligned} d = & 247189877625772708082305032045364310063716155961786944709 \\ & 001304405442364141942650918239813133509302592076274093052 \\ & 279652452641774620280825414234940684665038510519922304362 \\ & 234099709432670635240551383589379951595365548628008052316 \\ & 4421309842252041083571481397311251010410651998365589300540 \\ & 1938472951463059469690990532702482123571999423343020290770 \\ & 29792033908162989 \end{aligned}$$

4.2. Performance experiment

We conducted an experiment to estimate the total time cost of our attack in the special case where $n = 4$, i.e., $\varphi_n(N) = (p^4 - 1)(q^4 - 1)$, and the private exponent $d \approx \varphi_n(N) - N^\delta$. The results are presented in Table 1. Notice that the last column represents the average time it takes to obtain $p + q$.

Table 1: Experimental results for our attack with $\tau = 1$, $s = 6$, and $d \approx \varphi(N) - N^\delta$.

Modulus Bitsize	n	δ	m	Number of Instances	Total Average Time Cost (seconds)
100	4	0.8	3	1000	0.18
256	4	0.9	3	700	0.21
512	4	0.8	3	500	0.44
1024	4	0.9	3	200	1.31
2048	4	0.7	3	100	4.50
4096	4	0.7	3	10	25.00
1024	1	0.2	3	500	0.08
1024	2	0.4	3	400	0.26
1024	3	0.6	3	100	1.23
1024	5	1.0	3	60	9.50

Continued on next page

Table 1 – Continued from previous page

Modulus Bitsize	n	δ	m	Number of Instances	Total Average Time Cost (seconds)
1024	6	1.2	3	50	5.35
1024	7	1.4	3	30	50.30
1024	8	1.6	3	30	19.24
1024	9	1.8	3	10	278
1024	10	2	3	10	67.63

5. Our attack vs Existing attacks

In this section, we compare our attack to the existing attacks against the Cotan and Teşeleanu cryptosystem [8].

5.1. Our attack vs the attack of Teşeleanu [26]

Recently, Teşeleanu [26] has proposed a lattice attack against the scheme of Cotan and Teşeleanu cryptosystem [8]. He showed that if $q < p < 2q$, $d < N^\delta$ and $e = N^\beta$, then one can factor N in polynomial time if

$$\begin{cases} \delta \leq n - \sqrt{0.5\beta n} & \text{if } \frac{n}{2} \leq \beta \leq \frac{(n+1)^2}{2n}, \\ \delta \leq \frac{3n-1}{4} - \frac{n\beta}{2(n+1)} & \text{if } \frac{(n+1)^2\theta}{n} \leq \beta \leq \frac{(n+1)(3n-1)}{2n}. \end{cases}$$

Some of the differences between his attack and ours are as follows:

- In Theorem 7 and Theorem 8, if we set $\theta = \frac{1}{2}$ and neglect γ , i.e., set $\gamma = 0$, we obtain exactly his bound.
- His attack deals only with small private exponents d , whereas ours is valid for both small and large d .
- His attack is supposed to work for balanced moduli $N = pq$, i.e., $q < p < 2q$, whereas ours is valid for arbitrary primes p and q . Moreover, in the proof of its main result, Teşeleanu [26] assumes that $N^n < \varphi_n(N) = (p^n - 1)(q^n - 1)$, which is not the case. We correct this by showing that there exists a small integer s such that $\frac{N^n}{s} < \varphi_n(N)$.

5.2. Our attack vs the attack of Cotan and Teşeleanu [8]

Cotan and Teşeleanu [8] studied an attack based on continued fractions on their cryptosystem. They showed that if e and d satisfy the key equation $ed - k(p^n - 1)(q^n - 1) = 0$ and d is too small (i.e., d satisfies Equation 1), then one can factor N in polynomial time. Note that their attack only deals with small private exponents, whereas our lattice attack is valid for both small and large private exponents d . Furthermore, when $e \approx N^n$, they showed that if $d < N^{\frac{n}{4}}$ (i.e., $\delta < \frac{n}{4}$), then p and q can be recovered in polynomial time. For our attack, when $e \approx N^n$ and we take $\theta = \frac{1}{2}$ and $\gamma \approx 0$ in Theorem 7, the vulnerable bound for d is

$$\delta < \frac{n+1}{4} - \frac{1}{2(n+1)} \quad \text{for } n > 2.$$

Notice that

$$\frac{n}{4} < \frac{n+1}{4} - \frac{1}{2(n+1)} \quad \text{for } n > 2,$$

This show that our bound is better.

5.3. Our attack vs the attack of Seck et al. [24]

Combining continued fractions and Coppersmith's algorithm, Seck *et al.* [24] have recently showed that for an RSA modulus $N = pq$ with $q < p < 2q$, if the private exponent d satisfies the following bounds

$$d < \sqrt{\frac{2N^4 - 49N^2 + 2}{4N + 170N^2}} \text{ or } |\varphi_4(N) - d| < \sqrt{\frac{2N^4 - 49N^2 + 2}{4N + 170N^2}}$$

where $\varphi_4(N) = (p^4 - 1)(q^4 - 1)$, then one can find p and q in polynomial time in $\log(N)$. Notice this attack only concerns the particular case of $n = 4$, whereas our attack is valid for any $n \geq 1$.

6. Conclusion

In this paper, we have studied the general key equation $eu - (p^n - 1)(q^n - 1)v = \pm 1$ where e is a public exponent in the RSA variant of Cotan and Teşeleanu [8]. We have successfully extended the lattice attack of Teşeleanu [26] by showing that if the private exponent d is either too large or too small, then the modulus N can be factored in polynomial time using lattice reduction techniques and Coppersmith's method for solving modular multivariate polynomial equations.

References

- [1] Hatem M. Bahig, Dieaa I. Nassr, Mohammed A. Mahdi, and Hazem M. Bahig. Small private exponent attacks on RSA using continued fractions and multicore systems. *Symmetry*, 14(9), 2022. ISSN 2073-8994. doi: 10.3390/sym14091897. URL <https://www.mdpi.com/2073-8994/14/9/1897>.
- [2] Johannes Blömer and Alexander May. A generalized Wiener attack on RSA. In *International Workshop on Public Key Cryptography*, pages 1–13. Springer, 2004.
- [3] Dan Boneh. Twenty years of attacks on the RSA cryptosystem. *Notices of the AMS*, 46(2):203–213, 1999.
- [4] Dan Boneh and Glenn Durfee. Cryptanalysis of RSA with private key d less than $n^{0.292}$. In *Advances in Cryptology—EUROCRYPT’99: International Conference on the Theory and Application of Cryptographic Techniques Prague, Czech Republic, May 2–6, 1999 Proceedings 18*, pages 1–11. Springer, 1999.
- [5] Martin Bunder, Abderrahmane Nitaj, Willy Susilo, and Joseph Tonien. A generalized attack on RSA type cryptosystems. *Theoretical Computer Science*, 704:74–81, 2017.
- [6] Don Coppersmith. Finding a small root of a univariate modular equation. In *International Conference on the Theory and Applications of Cryptographic Techniques*, pages 155–165. Springer, 1996.
- [7] Don Coppersmith. Finding a small root of a bivariate integer equation; factoring with high bits known. In *International Conference on the Theory and Applications of Cryptographic Techniques*, pages 178–189. Springer, 1996.
- [8] Paul Cotan and George Teşeleanu. Small private key attack against a family of RSA-like cryptosystems. In *Nordic Conference on Secure IT Systems*, pages 57–72. Springer, 2023.
- [9] Koen de Boer and Wessel van Woerden. Lattice-based cryptography: A survey on the security of the lattice-based nist finalists. *Cryptology ePrint Archive*, 2025.

- [10] B. de Weger. Cryptanalysis of RSA with small prime difference. *Appl. Algebra Eng. Commun. Comput.*, 13(1):17–28, 2002.
- [11] The Sage Developers. *SageMath, the Sage Mathematics Software System (Version 10.4)*. SageMath, <https://www.sagemath.org>, 2024.
- [12] H Elkamchouchi, K Elshenawy, and H Shaban. Extended RSA cryptosystem and digital signature schemes in the domain of gaussian integers. In *The 8th International Conference on Communication Systems, 2002. ICCS 2002.*, volume 1, pages 91–95. IEEE, 2002.
- [13] Julien Grieru, Fabien Laguillaumie, and Thomas Pornin. Improving small private exponent attack on the Murru-Saettone cryptosystem. *Theoretical Computer Science*, 925:61–76, 2022. doi: 10.1016/j.tcs.2022.05.010. URL <https://doi.org/10.1016/j.tcs.2022.05.010>.
- [14] Johan Håstad. On using RSA with low exponent in a public key network. In *Advances in Cryptology – CRYPTO ’85*, volume 218 of *Lecture Notes in Computer Science*, pages 403–408. Springer, 1986. doi: 10.1007/3-540-47721-7_29.
- [15] Nicholas Howgrave-Graham. Finding small roots of univariate modular equations revisited. In *IMA International Conference on Cryptography and Coding*, pages 131–142. Springer, 1997.
- [16] Noboru Kunihiro. On optimal bounds of small inverse problems and approximate gcd problems with higher degree. In *International Conference on Information Security*, pages 55–69. Springer, 2012.
- [17] Arjen K Lenstra, Hendrik Willem Lenstra, and László Lovász. Factoring polynomials with rational coefficients. *Mathematische annalen*, 261:515–534, 1982.
- [18] Rudolf Lidl and Harald Niederreiter. *Finite Fields*, volume 20 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 2nd edition, 1997. ISBN 9780521561996.
- [19] Alexander May. *New RSA vulnerabilities using lattice reduction methods. Ph.D. Thesis, University of Paderborn, Paderborn, Germany*. PhD thesis, Citeseer, 2003.

- [20] Dieaa I. Nassr, M. Anwar, and Hatem M. Bahig. Improving small private exponent attack on the murru-saettone cryptosystem. *Theoretical Computer Science*, 923:222–234, 2022. ISSN 0304-3975. doi: <https://doi.org/10.1016/j.tcs.2022.05.010>. URL <https://www.sciencedirect.com/science/article/pii/S0304397522003085>.
- [21] Abderrahmane Nitaj. Another generalization of Wiener’s attack on RSA. In *International Conference on Cryptology in Africa*, pages 174–190. Springer, 2008.
- [22] R Rivest, A Shamir, and L Adleman. A method for obtaining digital signatures and public-key cryptosystems. *Communications of the ACM*, 1978.
- [23] Aymen Ben Sassi, Sami Mnasri, and Mounir Maller. Small private exponent attacks on RSA using continued fractions and multicore systems. *Symmetry*, 14(9):1897, 2022. doi: 10.3390/sym14091897. URL <https://doi.org/10.3390/sym14091897>.
- [24] Michel Seck, Oumar Niang, Djiby Sow, Abderrahmane Nitaj, Mengce Zheng, and Maher Boudabra. A new generalized attack on RSA-like cryptosystems. In Abderrahmane Nitaj, Svetla Petkova-Nikova, and Vincent Rijmen, editors, *Progress in Cryptology – AFRICACRYPT 2025*, pages 126–144, Cham, 2026. Springer Nature Switzerland. ISBN 978-3-031-97260-7. doi: 10.1007/978-3-031-97260-7_7. URL https://doi.org/10.1007/978-3-031-97260-7_7.
- [25] Joseph Surin and Shaanan Cohney. A gentle tutorial for lattice-based cryptanalysis. Cryptology ePrint Archive, Paper 2023/032, 2023. URL <https://eprint.iacr.org/2023/032>.
- [26] George Teşeleanu. A lattice attack against a family of RSA-like cryptosystems. In *International Symposium on Cyber Security, Cryptology, and Machine Learning*, pages 343–355. Springer, 2024.
- [27] Michael J Wiener. Cryptanalysis of short RSA secret exponents. *IEEE Transactions on Information theory*, 36(3):553–558, 1990.